

Exercises: Decision trees

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1 Exercise 1

Let consider a sample of data with $n = 8$ observations to predict the customer churn on the basis of 3 covariates. The description of the variables is provided in the table below.

Attribute	Description	Values
Gender	Gender.	{Male, Female, Other}
Unlimited Data	Unlimited data subscription (monthly extra fee).	{Yes, No}
Phone Service	Subscription to home phone service with the company.	{Yes, No}
Churn	If the customer left the company.	{Yes, No}

For your analysis you have the following manually classified training examples:

ID	Gender	Unlimited Data	Phone Service	Churn
1	Male	Yes	Yes	Yes
2	Female	Yes	No	No
3	Other	No	No	Yes
4	Male	No	No	No
5	Other	Yes	No	No
6	Male	Yes	No	No
7	Female	No	Yes	Yes
8	Female	Yes	Yes	Yes

a) Construct a decision tree based on this training data. For splitting, use cross-entropy as measure for impurity. Build a separate branch for each attribute. The decision tree shall stop when all instances in the branch have the same class, you do not need to apply a pruning algorithm.

Solution a) Construct a decision tree using cross-entropy

Step 1: Initial Entropy

$$\begin{aligned} H(R) &= - \left(\frac{4}{8} \log_2 \frac{4}{8} + \frac{4}{8} \log_2 \frac{4}{8} \right) \\ &= -(0.5 \times -1 + 0.5 \times -1) = 1 \end{aligned}$$

Step 2: Information Gain Calculations

We shall use the Information Gain to find the splitting variable in the first split.

Let R the region and $A \in \mathcal{A}$ a splitting variables, the Information Gain IG can be written as

$$IG(R, A) = H(R) - \sum_{v \in \mathcal{A}} \frac{|R_v|}{|R|} H(R_v)$$

Gender

$$\begin{aligned} IG(R, \text{Gender}) &= 1 - \left(\frac{3}{8} \times \left(-\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \right) + \frac{3}{8} \times \left(-\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) \right) \right. \\ &\quad \left. + \frac{2}{8} \times \left(-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \right) = 1 - \left(\frac{3}{8} \times 0.918 + \frac{3}{8} \times 0.918 + \frac{2}{8} \right) \\ &\approx 0.06 \end{aligned}$$

Unlimited Data

$$\begin{aligned} IG(R, \text{Unlimited Data}) &= 1 - \left(\frac{5}{8} \times \left(-\frac{2}{5} \log_2 \left(\frac{2}{5} \right) - \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right) + \right. \\ &\quad \left. \frac{3}{8} \times \left(-\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right) \right) \right) \\ &= 1 - \left(\frac{5}{8} \times 0.971 + \frac{3}{8} \times 0.918 \right) \approx 0.05 \end{aligned}$$

Phone Service

$$\begin{aligned} IG(R, \text{Phone Service}) &= 1 - \left(\frac{5}{8} \times \left(-\frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \frac{4}{5} \log_2 \left(\frac{4}{5} \right) \right) + \frac{3}{8} \times (-1 \log_2 (1)) \right) \\ &= 1 - \left(\frac{5}{8} \times 0.722 \right) \approx 0.55 \end{aligned}$$

The best splitting variable is **Phone service**. For the branch Phone Service = Yes, we don't need to further split the tree while we need to further split the tree when that condition is false.

Then, we should find another split variable for the following subset of data

ID	Gender	Unlimited Data	Phone Service	Churn
2	Female	Yes	No	No
3	Other	No	No	Yes
4	Male	No	No	No
5	Other	Yes	No	No
6	Male	Yes	No	No

$$H(R_{PS}) = - \left(\frac{4}{5} \log_2 \frac{4}{5} + \frac{1}{5} \log_2 \frac{1}{5} \right) \approx 0.72$$

Gender

$$\begin{aligned} IG(R_{PS}, \text{Gender}) &= 0.72 - \left(\frac{1}{5} \times (-1) \log_2(1) + \frac{2}{5} \times \left(-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \right. \\ &\quad \left. + \frac{2}{5} \times (-1 \log_2(1)) \right) = 0.72 - \left(\frac{1}{5} \times 0 + \frac{2}{5} \times 1 + \frac{2}{5} \times 0 \right) \approx 0.72 - 0.4 = 0.32 \end{aligned}$$

Unlimited Data

$$\begin{aligned} IG(R_{PS}, \text{Unlimited Data}) &= 0.72 - \left(\frac{3}{5} \times (-1) \log_2(1) + \frac{2}{5} \times \left(-\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \right) \\ &= 0.72 \left(\frac{3}{5} \times 0 + \frac{2}{5} \times 1 \right) \approx 0.72 - 0.4 = 0.32 \end{aligned}$$

Resulting Tree Structure:

Phone Service

Yes → "Yes" (IDs 1,7,8)

No

Unlimited Data

Yes → "No" (IDs 2,5,6)

No

Gender: Other → "Yes" (ID 3)

Gender: Female/Male → "No" (ID 4)

b) Apply the decision tree to the following customers:

ID	Gender	Unlimited Data	Phone Service
9	Male	No	No
10	Female	Yes	Yes
11	Male	No	Yes

Solution b) Predictions for New Customers

ID	Gender	Unlimited Data	Phone Service	$\hat{Churn}$
9	Male	No	No	No
10	Female	Yes	Yes	Yes
11	Male	No	Yes	Yes

c) We receive data about two new customers

ID	Gender	Unlimited Data	Phone Service
12	Male	Yes	No
13	Female	No	Yes

and, after few years, the manager provided us the actual outcomes

ID	Churn
9	Yes
10	Yes
11	Yes
12	No
13	No

Please, propose and compute metrics providing some comments.

Solution c) Performance Metrics

The predicted responses are

ID	$\hat{Churn}$
9	No
10	Yes
11	Yes
12	No
13	Yes

Confusion Matrix:

		Predicted	
		No	Yes
Actual	No	1 (ID12)	1 (ID13)
	Yes	1 (ID9)	2 (ID10,11)

$$\text{Accuracy} = \frac{3}{5} = 0.6$$

$$\text{Precision} = \frac{2}{3} \approx 0.667$$

$$\text{Recall} = \frac{2}{3} \approx 0.667$$

$$\text{F1-Score} = 0.667$$

We don't have enough data to evaluate the model. Indeed, we use few observations in the training set to learn the model while using 5 observations to evaluate it. In general, in this kind of situation we should avoid to make comments. However, discarding this consideration and evaluating just the values obtained from the metrics, we can say that the accuracy is around the middle. In classical cases, these kind of values should bring us to improve the model.

d) Now, try to use the missclassification rate as splitting criterion using the same training data. Provide some comments and compare results with the previous case.

Solution d) Misclassification Rate Comparison

Let consider that $Y \in \mathcal{Y} = \{0, 1\}$ where $Y = Churn$, we can compute the missclassification rate of the first region

$$\mathcal{J}(R) = 1 - \max_{Y \in \mathcal{Y}}(p_Y^R) = 0.5$$

Gender

$$\begin{aligned} IG(R, \text{Gender}) &= 0.5 - \left(\frac{3}{8} \times 0.33\right) + \left(\frac{3}{8} \times 0.33\right) + \left(\frac{2}{8} \times 0.5\right) \\ &= 0.5 - (0.124 + 0.124 + 0.125) \approx 0.127 \end{aligned}$$

Unlimited Data

$$\begin{aligned} IG(R, \text{Unlimited Data}) &= 0.5 - \left(\frac{5}{8} \times 0.4\right) + \left(\frac{3}{8} \times 0.33\right) \\ &= 0.5 - (0.25 + 0.124) \approx 0.126 \end{aligned}$$

Phone Service

$$\begin{aligned} IG(R, \text{Phone Service}) &= 0.5 - \left(\frac{5}{8} \times 0.2\right) + \left(\frac{3}{8} \times 0\right) \\ &= 0.5 - (0.125) \approx 0.375 \end{aligned}$$

The first splitting variable is still **Phone Service**. For the branch Phone Service = Yes, we don't need to further split the tree while we need to further split the tree when that condition is false.

Then, we should find another split variable for the following subset of data

ID	Gender	Unlimited Data	Phone Service	Churn
2	Female	Yes	No	No
3	Other	No	No	Yes
4	Male	No	No	No
5	Other	Yes	No	No
6	Male	Yes	No	No

$$\mathcal{J}(R_{PS}) = 1 - \max_{Y \in \mathcal{Y}}(p_Y^{R_{PS}}) = 0.2$$

Gender

$$\begin{aligned} IG(R, \text{Gender}) &= 0.2 - \left(\frac{1}{2} \times 0\right) + \left(\frac{2}{5} \times 0.5\right) + \left(\frac{2}{5} \times 0\right) \\ &= 0.5 - 0.2 \approx 0.3 \end{aligned}$$

Unlimited Data

$$\begin{aligned} IG(R, \text{Unlimited Data}) &= 0.2 - \left(\frac{3}{5} \times 0\right) + \left(\frac{2}{5} \times 0.5\right) \\ &= 0.5 - 0.2 \approx 0.3 \end{aligned}$$

In this case we obtain the same result. Then, the tree corresponds to **Resulting Tree Structure**:

Phone Service

Yes → "Yes" (IDs 1,7,8)

No

Unlimited Data

Yes → "No" (IDs 2,5,6)

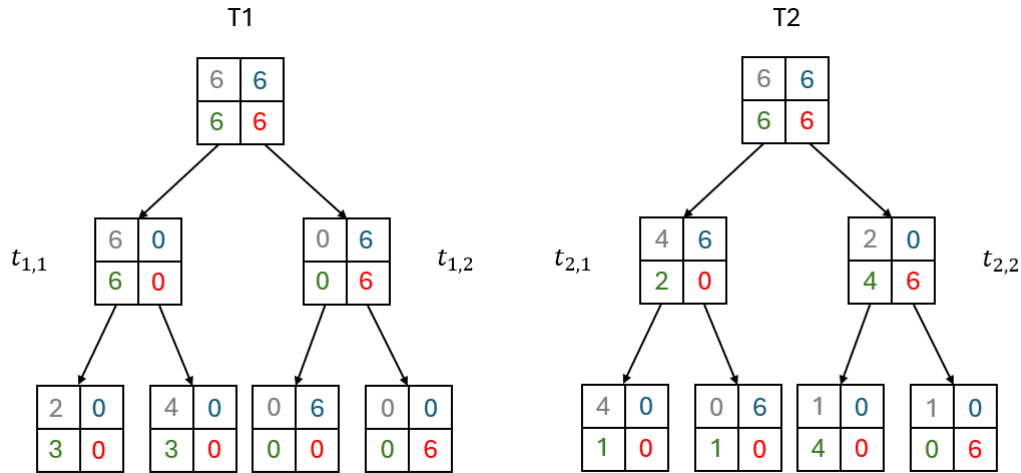
No

Gender: Other → "Yes" (ID 3)

Gender: Female/Male → "No" (ID 4)

2 Exercise 2

Let $\mathcal{D}$ be a set of examples over a feature space X and the outcome space $Y = \{c_1, c_2, c_3, c_4\}$, with $|\mathcal{D}| = 24$. Consider the following illustration of two possible decision trees, T_1 and T_2 - the coloured numbers represent the absolute frequency of each classes of Y in each node $t_{i,j}$, where $i = \{1, 2\}$ indicates the tree number and $j = \{1, 2, 3, 4, 5, 6\}$ is the node number.



a) First, consider only the first split that each of the two trees makes: compute the missclassification rate and the cross-entropy criterions of splits $t_{1,1}$, $t_{1,2}$, $t_{2,1}$, $t_{2,2}$. Interpret the results: which first split is better among the two decision trees?

Solution a) Compare First Splits

Missclassification rate

$$\mathcal{J}(R_{t_1}) = \frac{12}{24}0.5 + \frac{12}{24}0.5$$

$$\mathcal{J}(R_{t_2}) = \frac{12}{24}0.5 + \frac{12}{24}0.5$$

Cross-Entropy

$$H(R_{t_1}) = -\frac{12}{24} \left(\frac{1}{2} \log_2(0.5) + \frac{1}{2} \log_2(0.5) \right) - \frac{12}{24} \left(\frac{1}{2} \log_2(0.5) + \frac{1}{2} \log_2(0.5) \right) = 2$$

$$\begin{aligned}
H(R_{t_2}) &= -\frac{12}{24} \left(\frac{4}{12} \log_2 \left(\frac{4}{12} \right) + \frac{2}{12} \log_2 \left(\frac{2}{12} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) \\
&\quad - \frac{12}{24} \left(\frac{4}{12} \log_2 \left(\frac{4}{12} \right) + \frac{2}{12} \log_2 \left(\frac{2}{12} \right) + \frac{1}{2} \log_2 \left(\frac{1}{2} \right) \right) = 2.92
\end{aligned}$$

We observe that the misclassification rate yields the same impurity value (0.5) across all data subsets, regardless of the class distribution. This is because the misclassification rate is a coarse measure—it only considers the majority class in each subset and ignores how the remaining data is distributed among the minority classes.

In contrast, cross-entropy produces varying impurity values and is more sensitive to the degree of class mix within each subset. For example, the subsets of T_2 receive a higher entropy value (1.46), indicating a more mixed and uncertain distribution of classes.

This distinction explains why cross-entropy tends to produce better splits early in the tree. It penalizes uncertain or heterogeneous splits more heavily, leading to purer subsets. Meanwhile, the misclassification rate may overlook these nuances and result in suboptimal early splits.

b) If we compare T_1 and T_2 in terms of their misclassification rate on D , which one is the best decision tree?

Solution b) Tree Comparison

Missclassification rate

- T_1 : $\mathcal{J}(T_1) = \frac{5}{12}0.4 + \frac{7}{12}0.43 + 0 + 0 = 0.35$
- T_2 : $\mathcal{J}(T_2) = \frac{5}{12}0.2 + \frac{7}{12}0.143 + \frac{5}{12}0.2 + \frac{7}{12}0.143 = 0.33$

Cross-entropy

- T_1 : $H(T_1) = -\frac{5}{12} \left(\frac{2}{5} \log_2 \left(\frac{2}{5} \right) + \frac{3}{5} \log_2 \left(\frac{3}{5} \right) \right) - \frac{7}{12} \left(\frac{4}{7} \log_2 \left(\frac{4}{7} \right) + \frac{3}{7} \log_2 \left(\frac{3}{7} \right) \right) = 0.98$
- T_2 : $H(T_2) = -\frac{5}{12} \left(\frac{4}{5} \log_2 \left(\frac{4}{5} \right) + \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right) - \frac{7}{12} \left(\frac{6}{7} \log_2 \left(\frac{6}{7} \right) + \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right) - \frac{5}{12} \left(\frac{4}{5} \log_2 \left(\frac{4}{5} \right) + \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right) - \frac{7}{12} \left(\frac{6}{7} \log_2 \left(\frac{6}{7} \right) + \frac{1}{7} \log_2 \left(\frac{1}{7} \right) \right) = 1.29$

The missclassification rate prefers T_1 while the cross-entropy prefers T_2 .